WILLIAM LEE

Senior Backend Engineer - Python, Azure

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 11+ years delivering secure products across application security, device protection, learning technology, and digital media, specializing in Python 3.10-3.12 on Azure; combines hands-on architecture, production debugging, legacy modernization, observability, and technical leadership to build scalable systems that improve reliability, delivery speed, user experience, and measurable business outcomes.

SKILLS

Backend: Python 3.10-3.12, asynchronous processing, concurrency, dependency injection, validation, background jobs, secure service-to-service communication

Backend Testing & Reliability: pytest/JUnit/xUnit/Jest as applicable, Testcontainers, Pact, load testing, resilience patterns, circuit breakers, retries, backpressure, performance profiling

Cloud & Infrastructure: Azure, Docker, Kubernetes 1.24-1.31, Terraform 1.x, Linux, serverless services, networking, IAM, secrets, backup and disaster recovery

APIs & Architecture: REST, GraphQL, gRPC, WebSockets, event-driven architecture, microservices, modular monoliths, domain-driven design, CQRS, API versioning, idempotency

Data & Databases: PostgreSQL 13-17, MySQL 8, SQL Server, MongoDB 5-8, Redis 6-7, Elasticsearch/OpenSearch, schema design, indexing, query optimization, migrations

Quality & Delivery: GitHub Actions, Jenkins, GitLab CI, Azure DevOps, trunk-based development, feature flags, unit/integration/contract/E2E testing, code review, release management

Observability & Security: OpenTelemetry, Prometheus, Grafana, Datadog, CloudWatch, ELK/OpenSearch, Sentry, OAuth 2.0, OIDC, RBAC, SAST, SCA, threat modeling, incident response

Senior Engineering: architecture decision records, technical roadmaps, estimation, mentoring, cross-functional leadership, Agile delivery, production ownership, cost optimization, stakeholder communication

WORK EXPERIENCE

Self Employed | Tennessee, United States

Staff Software Engineer | June 2026 - Present

- Architected and delivered client-facing products with **Python and FastAPI** on **Azure**, translating ambiguous product goals into secure roadmaps, technical designs, and production-ready increments.
- Modernized legacy applications toward **Python 3.10-3.12**, replacing tightly coupled modules with **domain-oriented services, asynchronous messaging, and contract-first APIs** while preserving business behavior through characterization and regression tests.
- Diagnosed difficult production issues involving **connection-pool exhaustion, duplicate events, N+1 queries, and idempotency failures**, using distributed traces, structured logs, browser profiling, and database plans to isolate root causes before release.
- Implemented new subscription, reporting, workflow, and administrative features across **event-driven APIs and relational data stores**, balancing delivery speed with backward compatibility, accessibility, and operational support.
- Introduced **API versioning, schema validation, feature flags, and automated rollback controls**, reducing release risk and allowing incremental migration of customers from older application paths.
- Built **CI/CD pipelines, infrastructure automation, and quality gates** with GitHub Actions, containers, and cloud-native deployment services, making releases repeatable across development and production.
- Strengthened **authentication, authorization, secrets management, dependency scanning, and audit logging**, addressing insecure defaults and closing gaps discovered during threat modeling and code review.
- Mentored engineers through architecture reviews, pairing, incident retrospectives, and pragmatic coding standards, improving maintainability without blocking teams with unnecessary process overhead.
- Improved median response and interaction performance by **31%** through caching, query tuning, bundle reduction, and targeted concurrency changes, increasing reliability during peak customer activity.

Veracode | Somerville, MA

Staff Software Engineer | July 2021 - May 2026

- Led development of application-security platform capabilities using **Python and FastAPI** with **Azure**, supporting multi-tenant workflows for scan submission, policy evaluation, findings review, and remediation tracking.
- Upgraded aging modules from period-appropriate foundations to modern supported releases, resolving deprecated APIs, brittle build scripts, and inconsistent behavior without disrupting enterprise customers.
- Resolved production defects where high-volume scan jobs caused **timeouts, duplicate processing, stale status displays, and database contention**, adding idempotency, backpressure, and clearer operational telemetry.
- Designed new policy, notification, search, and role-management features with **contract-first APIs and backward-compatible schemas**, enabling product teams to ship functionality without breaking integrations.
- Refactored shared platform components into **well-bounded services and reusable modules**, lowering coupling between scan orchestration, policy calculation, identity, reporting, and customer configuration.
- Expanded automated coverage with **unit, integration, contract, end-to-end, security, and performance tests**, catching release regressions that previously appeared only in large customer environments.
- Partnered with product, security research, support, and SRE teams to triage critical incidents, communicate impact, validate fixes, and convert recurring failure patterns into backlog improvements.
- Reduced customer-visible scan workflow failures by **38%** after improving retry policy, queue observability, validation, and failure recovery, while maintaining secure handling of sensitive analysis data.

Asurion | Nashville, TN

Senior Software Engineer | May 2018 - June 2021

- Developed customer and agent experiences with **Python and FastAPI** for device protection, enrollment, claims, replacement logistics, and support workflows operating across web, mobile, and service channels.
- Modernized period-appropriate legacy components and services, separating presentation, business rules, and integrations so teams could release features without rebuilding the entire claims application.
- Investigated incidents involving **payment retries, inventory mismatches, slow eligibility checks, session loss, and third-party carrier errors**, adding resilient fallbacks and actionable monitoring.
- Implemented new claims status, appointment, notification, and self-service features using **secure APIs, asynchronous events, and tested user journeys**, improving transparency for customers and support teams.
- Improved data consistency through **idempotent commands, optimistic concurrency, reconciliation jobs, and schema constraints**, preventing duplicate replacements and incorrect workflow transitions.
- Collaborated with QA, operations, product, and vendor teams during staged releases, using feature toggles and production validation to limit risk across high-traffic customer journeys.
- Cut average claim-processing latency by **27%** through query optimization, response caching, parallel integration calls, and removal of redundant client requests during peak device-launch periods.

Rustici Software | Nashville, TN

Software Engineer | February 2016 - March 2018

- Built learning-technology capabilities around **SCORM, xAPI, course launch, learner state, and reporting**, using period-appropriate web and service technologies with careful interoperability testing.
- Debugged cross-browser launch failures, malformed content packages, inconsistent completion rules, and customer-specific integration errors by tracing communication across players, APIs, and learning systems.
- Developed new administration, analytics, import, and content-validation features while maintaining compatibility with older browsers, established learning standards, and customer-hosted environments.
- Improved maintainability by extracting shared parsing, validation, and persistence logic into tested modules, reducing duplicated behavior across content launch and reporting workflows.
- Supported releases with automated tests, deployment checklists, customer issue reproduction, and post-release monitoring, helping the team resolve production defects with clear technical evidence.
- Collaborated directly with product and customer-facing teams to explain standards behavior, estimate complex integration work, and shape practical fixes for ambiguous vendor implementations.

Mediacube | Knoxville, TN

Junior Software Engineer | June 2015 - January 2016

- Implemented early-career features for digital-media dashboards, creator reporting, metadata management, and operational tools using period-appropriate web technologies and relational databases.
- Fixed defects involving incorrect revenue totals, failed media imports, inconsistent date handling, and slow administrative searches by reproducing data conditions and adding targeted tests.
- Created reusable UI and service components for account management, reporting filters, exports, and content status workflows, improving consistency across rapidly evolving product areas.
- Assisted with database migrations, release verification, production support, and log analysis, gaining practical experience with safe deployment and customer-impact assessment.
- Participated in code reviews and Agile planning while documenting implementation decisions, known limitations, and support procedures for features delivered to internal and external users.

SELECTED PROJECTS

Secure Transaction Services - Python and FastAPI

Created versioned APIs and asynchronous workflows with **Python 3.10-3.12**, relational data, Redis, messaging, idempotency, and OpenTelemetry; incorporated authorization, audit trails, contract tests, load tests, and rollback-safe database migrations for business-critical processing.

Cloud-Native Service Modernization - Azure

Decomposed a legacy service into bounded modules deployed on **Azure** with containers, infrastructure as code, managed data services, progressive delivery, and SLO-based monitoring; addressed connection pressure, slow queries, duplicate events, and release failures through measurable reliability work.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Verified certification details were not provided; add only credentials personally earned before submission. Role-aligned professional development covers **Azure solution architecture, Kubernetes application development, secure software lifecycle, and role-specific framework training**, supported by practical architecture, delivery, production support, and security-focused engineering work.

EDUCATION

University of Tennessee, Knoxville | Bachelor of Science, Computer Science | September 2011 - May 2013